

$$1) \lim_{x \rightarrow 0} \frac{2 \tan x + x}{x} = \lim_{x \rightarrow 0} 2 \left(\frac{\tan x}{x} \right) + \frac{x}{x} = 2 + 1 = 3$$

$$2) \lim_{x \rightarrow 0} \frac{\sin x + 3x}{x + 2 \sin x} = \frac{\cancel{x} \left(\frac{\sin x}{x} + 3 \right)}{\cancel{x} \left(1 + 2 \left(\frac{\sin x}{x} \right) \right)} = \frac{6}{3} = 2$$

$$3) \lim_{x \rightarrow 0} \frac{2x \sin x}{\tan^2 x} = \lim_{x \rightarrow 0} \frac{2x \cancel{\sin x}}{\frac{\sin^2 x}{\cos^2 x}} = \lim_{x \rightarrow 0} \frac{2x \cos^2 x}{\sin x} = 2 \cdot 1 \cdot 1 = 2$$

$$4) \lim_{x \rightarrow 0} \frac{2 \sin x + 3x}{3 \sin x - x} = \lim_{x \rightarrow 0} \frac{\cancel{x} \left(2 \frac{\sin x}{x} + 3 \right)}{\cancel{x} \left(3 \frac{\sin x}{x} - 1 \right)} = \frac{7}{2}$$

$$5) \lim_{x \rightarrow 0} \frac{\sin x - 2x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} - \frac{2x}{x} = -1$$

$$6) \lim_{x \rightarrow 0} \frac{x^2}{\sin x} = \left(\lim_{x \rightarrow 0} x \cdot x \cdot \frac{1}{\sin x} \right) = \lim_{x \rightarrow 0} \frac{x}{\cancel{x}} = 0$$

$$7) \lim_{x \rightarrow 0} \frac{\sin 2x + x}{x + \sin x} = \left(\lim_{x \rightarrow 0} 2x \cdot \frac{1}{x} \cdot \frac{1}{2} + \lim_{x \rightarrow 0} x \cdot \frac{1}{x} \right) = \lim_{x \rightarrow 0} \frac{2x + x}{x + x} = \frac{3x}{2x} = \frac{3}{2}$$

$$8) \lim_{x \rightarrow 0} \frac{x^2 + x}{2x + \sin x} = \lim_{x \rightarrow 0} \frac{\cancel{x} (x + 1)}{\cancel{x} \left(2 + \frac{\sin x}{x} \right)} = \frac{1}{3}$$

$$9) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{1 - \sin x} = \left(t = x - \frac{\pi}{2} \right) = \lim_{t \rightarrow 0} \frac{\cos(t + \frac{\pi}{2})}{1 - \sin(t + \frac{\pi}{2})} = \lim_{t \rightarrow 0} \frac{-\sin(t)}{1 - \cos(t)} = \text{NON ESISTE}$$

$$= \left(\lim_{t \rightarrow 0} t \cdot \frac{1}{t} \cdot \frac{1}{1 - \cos t} \right) = \lim_{t \rightarrow 0} \frac{-t}{\frac{1}{2} t^2} = \lim_{t \rightarrow 0} -2 \cdot \frac{1}{t} \rightarrow \begin{matrix} \lim_{t \rightarrow 0^+} -\frac{2}{t} = -\infty \\ \lim_{t \rightarrow 0^-} \frac{2}{t} = +\infty \end{matrix}$$

$$10) \lim_{x \rightarrow +\infty} \left(\frac{x^2 - 2x + 1}{x^2 - 4x + 7} \right)^{\frac{x^2 - x + 3}{2x + 5}} = \lim_{x \rightarrow +\infty} \left(1 + \frac{2x - 6}{x^2 - 4x + 7} \right)^{\frac{x^2 - x + 3}{2x + 5}} = e$$

$$= \lim_{x \rightarrow +\infty} \left(\frac{x^2 - 2x + 1 - 4x + 7 + 4x - 7}{x^2 - 4x + 7} \right)^{\frac{x^2 - x + 3}{2x + 5}} = \lim_{x \rightarrow +\infty} \left(1 + \frac{2x - 6}{x^2 - 4x + 7} \right)^{\frac{x^2 - x + 3}{2x + 5}} =$$

$$= \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{\left(\frac{x^2 - 4x + 7}{2x - 6} \right)} \right)^{\frac{x^2 - x + 3}{2x + 5} \cdot \frac{x^2 - 4x + 7}{2x - 6} \cdot \frac{2x - 6}{x^2 - 4x + 7}} =$$

$$= \lim_{x \rightarrow +\infty} \left(\left(1 + \frac{1}{\left(\frac{x^2 - 4x + 7}{2x - 6} \right)} \right)^{\frac{x^2 - 4x + 7}{2x - 6}} \right)^{\lim_{x \rightarrow +\infty} \left(\frac{x^2 - x + 3}{2x + 5} \cdot \frac{2x - 6}{x^2 - 4x + 7} \right)} =$$

$$= \lim_{x \rightarrow +\infty} \left(\frac{x^2 - x + 3}{2x + 5} \cdot \frac{x^2 - 4x + 7}{2x - 6} \right) = 1 \Rightarrow \lim_{x \rightarrow +\infty} \left(\frac{x^2 - 2x + 1}{x^2 - 4x + 7} \right)^{\frac{x - x + 3}{2x + 5}} = e$$

$$\begin{aligned} 11) \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x^x} \right)^{(x+1)^x} &= \lim_{x \rightarrow +\infty} \left(\left(1 + \frac{1}{x^x} \right)^{x^x} \right)^{\frac{(x+1)^x}{x^x}} = e^e \\ &= \lim_{x \rightarrow +\infty} \left(\left(1 + \frac{1}{x^x} \right)^{x^x} \right)^{\left(\frac{x+1}{x} \right)^x} = \lim_{x \rightarrow +\infty} \left(\left(1 + \frac{1}{x^x} \right)^{x^x} \right)^{\left(1 + \frac{1}{x} \right)^x} = e^e \end{aligned}$$

$$\begin{aligned} 12) \lim_{x \rightarrow +\infty} \left(\frac{x^4 + 5x^2 + 5}{(x^2 + 4)(x^2 + 2)} \right)^{x^2} &= \lim_{x \rightarrow +\infty} \left(\frac{\overbrace{x^4 + 5x^2 + 5}^{x^4 + 3x^2 + 2} + \overbrace{5 + 3x^2 + 2}^{2x^2 + 3}}{x^4 + 3x^2 + 2} \right)^{x^2} = \\ &= \lim_{x \rightarrow +\infty} \left(1 + \frac{2x^2 + 3}{x^4 + 3x^2 + 2} \right)^{x^2} = \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{\frac{x^4 + 3x^2 + 2}{2x^2 + 3}} \right)^{x^2} = \\ &= \lim_{x \rightarrow +\infty} \left(\underbrace{1 + \frac{1}{\frac{x^4 + 3x^2 + 2}{2x^2 + 3}}}_{e} \right)^{\frac{x^4 + 3x^2 + 2}{2x^2 + 3} \cdot x^2} = e^2 \end{aligned}$$

$\frac{2x^4 + 3x^2}{1x^4 + 3x + 2} = 2$

$$13) \lim_{x \rightarrow 0} \frac{\sin^2 x - 8^x + e^{-x}}{x} =$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{x} + \underbrace{\left(\frac{-8^x + 1}{x} \right)}_{\checkmark} + \frac{e^{-x} - 1}{x}$$

$$\left(\sin^2 x \sim x^2 \right) \Rightarrow \lim_{x \rightarrow 0} \frac{\sin^2(x)}{x} = \lim_{x \rightarrow 0} \frac{x^2}{x} = x = 0$$

$$\lim_{x \rightarrow 0} \frac{-8^x + 1}{x} = \lim_{x \rightarrow 0} - \left(\frac{8^x - 1}{x} \right) = -\log(8)$$

$$\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x} = \lim_{x \rightarrow 0} - \left(\frac{e^{-x} - 1}{-x} \right) = -1 \Rightarrow$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 x - 8^x + e^{-x}}{x} = 0 - \log(8) - 1 = -\log(8) - 1$$

$$14) \lim_{x \rightarrow 0^+} \frac{\sqrt{1 - \cos x}}{x} = \lim_{x \rightarrow 0^+} \sqrt{\frac{1 - \cos x}{x^2}} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$15) \lim_{x \rightarrow 0} \frac{1 - \cos x - \sin x}{x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \cdot x - \lim_{x \rightarrow 0} \frac{\sin x}{x} =$$

$$= 0 - 1 = -1$$

$$16) \lim_{x \rightarrow +\infty} \left(1 + \frac{3}{2\sqrt{x}}\right)^x = \lim_{x \rightarrow +\infty} \left(1 + \frac{3}{2\sqrt{x}}\right)^{\frac{1}{\frac{1}{2\sqrt{x}}} \cdot x} = e^{+\infty} = +\infty$$

$\lim_{x \rightarrow +\infty} \frac{x}{2\sqrt{x}} = +\infty$

$$17) \lim_{x \rightarrow 0} \frac{1 - \cos x - \sin x}{x^2} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} - \lim_{x \rightarrow 0} \frac{\sin x}{x^2} \cdot \left(\frac{1}{x}\right)$$

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$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$
 $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$

$$18) \lim_{x \rightarrow 0} \frac{\sin^2 2x}{x \tan x} = \lim_{x \rightarrow 0} \frac{\sin^2 2x}{x} \cdot \frac{1}{\tan x} = \left(\sin^2(2x) \sim (2x)^2 \text{ as } x \rightarrow 0 \right) =$$

$$= \lim_{x \rightarrow 0} \frac{4x^2}{x} \cdot \frac{\cos x}{\sin x} = \lim_{x \rightarrow 0} 4 \left(\frac{x}{\sin x} \cdot \cos x \right) = 4$$

$$19) \lim_{x \rightarrow \alpha} \frac{\sin x - \sin \alpha}{x - \alpha} = \left(\begin{matrix} t = x - \alpha \\ x = t + \alpha \\ \alpha = x - t \end{matrix} \right) = \lim_{t \rightarrow 0} \frac{\sin(t + \alpha) - \sin(\alpha)}{t} =$$

$$= \lim_{x \rightarrow 0} \frac{\sin(t) \cos(\alpha) + \cos(t) \sin(\alpha) - \sin(\alpha)}{t} =$$

$$= \lim_{x \rightarrow 0} \frac{\sin t \cos \alpha + \sin \alpha (\cos t - 1)}{t} = \cos \alpha \cdot \lim_{x \rightarrow 0} \frac{\sin t}{t} + \sin \alpha \cdot \lim_{x \rightarrow 0} \frac{\cos t - 1}{t}$$

$$\left(\frac{\cos t - 1}{t^2} \cdot t \right) = \cos \alpha \cdot 1 + 0 = \cos \alpha$$